

REAL TIME VEHICLE POLLUTION DETECTION AND TRACKING USING ITERATIVE DEEP LEARNING AND YOLO

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Abstract- As urbanization accelerates, the adverse impact of vehicular pollution on the environment becomes a pressing concern. In response, This article presents a novel deep learning-based framework for automatic vehicle pollution detection using on-road surveillance camera footage. In this model, designing and implementing a custom deep learning model along with the 7 deep learning models with improved accuracy and F1 score. Additionally, we extend the framework to process video data by integrating YOLOv5 for vehicle detection and OpenCV for image processing, enabling real-time pollution detection in traffic scenarios. Furthermore, we incorporate number plate tracking for identified polluting vehicles using Gemini AI and store the extracted information in a Firebase database for potential enforcement and accountability. To implement deep learning algorithms with a feedback based iterative model to enhance the accuracy and reliability of pollution detection. These enhancements significantly improve the accuracy, functionality, and real-world applicability of the framework, offering a robust and comprehensive solution for automatic vehicle pollution detection with significant implications for environmental monitoring and regulation.

Index Terms- Vehicle pollution detection, deep learning, transfer learning, iterative learning, real time surveillance.

I. INTRODUCTION

The peril of air pollution on public health, potentially leading to premature mortality, is a well-recognized concern [1]. Urban and peri-urban areas are particularly affected by this environmental hazard, primarily attributed to vehicle emissions, causing significant harm to the surrounding ecosystem [2]. Within the realm of Intelligent Transportation Systems (ITS), challenges persist, including the identification of polluting vehicles, advancement of emission-reducing technologies, monitoring highly polluted zones, and enforcement of traffic regulations. Swift, real-time detection is essential for timely interventions by Government Traffic Authorities.

Deep Learning (DL) approaches have gained substantial traction in recent times for addressing challenges in ITS [3], [4], [5]. Various methodologies have also been developed to identify and monitor air pollution from different perspectives [6], [7]. Notably, a deep spatio-

temporal-based fusion network was proposed for vehicle emission forecasting [8], and Xu et al. utilized an LSTM-based Autoencoder model for predicting vehicle pollution [9]. A wireless sensor network-based system for vehicle monitoring and air pollution detection was introduced [10], and Chandrasekaran et al. proposed an automated system within vehicles using sensors and microcontrollers [11]. Pal et al. designed a sensor-based system to share individual car pollution status [12], and a system using an MQ-7 carbon monoxide sensor for vehicle pollution identification was introduced [13]. However, these studies involve sensor installations, adding to overall costs.

In a different approach, Cardoso et al. pursued a US patent for an image-based technique assessing toxic gas absorption in vehicles [14]. Guo et al. presented empirical evidence showcasing a surge in electric vehicle sales due to escalating vehicle pollution [15]. However, the transition to 100% electric vehicles in third-world countries is anticipated to take at least another decade. Researchers have explored various image processing approaches [16], [17], [18]. Tao et al. proposed a Volume Local Binary Pattern (LBP)-based method for detecting smoky vehicles [19]. They also introduced a TAMURA feature-based technique for smoke detection over video [17]. Yuan et al. explored Higher Order Local Ternary Pattern (HLTP) for smoke detection [20], and a combination of LBP and Support Vector Machine (SVM) was employed for identifying smoke [21]. However, these methods may face challenges in scenarios involving overlapping smoke and non-smoke LBP.

In the domain of single-image-based deep learning approaches for vehicle smoke detection, few solutions exist. An approach utilizing a vision transformer for identifying smoky vehicles on roads was presented [22], while Ba et al. developed an attention-based CNN network for satellite smoke images [23]. Another approach employed a convolution network, DNCNN, for single-image smoke detection [24], with its performance closely tied to the fully connected network part of the convolution layers. Additional research solutions for smoke detection in diverse environments [25], [26] offer potential applicability in our specific problem domain.

Air pollution poses a significant threat to human health and the environment, with vehicle emissions being a major contributor. Identifying polluting vehicles is crucial for

enforcing regulations and improving air quality. While sensor-based methods exist, they often come with high installation and maintenance costs, making them less feasible for widespread deployment.

In this paper, we present a novel deep learning-based framework for automatic vehicle pollution detection using videos from roadside surveillance cameras. This approach leverages existing infrastructure, offering a cost-effective and scalable solution. Our work builds upon an existing framework that utilizes seven popular CNN models: Inception-V3, MobileNet-V2, MobileNet-V3, InceptionResNet-V2, VGG16, VGG19 and XceptionNet. These models are pre-trained on the ImageNet dataset and fine-tuned for the specific task of vehicle pollution detection. We further enhance this framework by introducing three key novelties.

- First, we design and implement a custom deep learning model specifically tailored for vehicle pollution detection. This model architecture incorporates 10 Conv2D filters, dropout layers, dense layers, and a flatten layer, and it demonstrates superior accuracy and F1 score compared to the pre-trained models used in the original framework.
- Second, we extend the framework to process video data, making it more practical for real-world applications. By integrating YOLOv5 for vehicle detection and OpenCV for image processing, the system analyzes individual frames and provides pollution detection results in video format. This enables real-time identification of polluting vehicles in traffic flows.
- Third, we incorporate number plate tracking for identified polluting vehicles using Gemini AI. The extracted number plate information, along with timestamps, is stored in a Firebase database. This facilitates potential enforcement measures and allows authorities to track and penalize violating vehicles.

To train and test the models, we have created a diverse dataset of 3000 vehicle images captured under various conditions, including variations in lighting, weather, and camera angles. This diversity ensures that the models are robust and can generalize well to real-world scenarios.

The core of the framework is an iterative deep learning process. Initially, the models are trained on a small subset of labeled images. Then, they are used to predict the pollution status of unlabeled images. Predictions with high confidence are added to the training set, and the models are re-trained on this expanded dataset. This feedback loop allows the models to learn from their own predictions and improve their accuracy over time.

To further enhance the reliability of the system, we employ a majority voting scheme. Instead of relying on the prediction of a single model, the final decision is made by combining the outputs of all seven models. This consensus approach reduces the impact of individual model errors and increases the overall confidence in the pollution detection results.

These enhancements significantly improve the accuracy, functionality, and real-world applicability of the framework. The custom model enhances prediction performance, video processing allows for real-time analysis of traffic scenarios, and number plate tracking with database integration facilitates the identification and potential penalization of violating vehicles. This work presents a robust and comprehensive solution for automatic vehicle pollution detection with significant implications for environmental monitoring and regulation.

The primary contributions of our proposed work can be summarized as follows:

1. Generic and Deployable Approach:

Our work introduces a novel and highly generic approach that can be seamlessly integrated into any supervised learning-based framework. This method proves particularly effective in scenarios where the supervised system is constrained by limited labeled data. Traditional methods often falter in such situations, while our proposed framework consistently demonstrates superior efficiency compared to these conventional approaches.

2. Enhanced Accuracy and Confidence:

One of the key achievements of our proposed approach is the substantial increase in both accuracy and confidence levels. Through various experiments, we observed a remarkable improvement in the confidence level of our model, a critical aspect often overlooked in traditional approaches. The incorporation of a majority voting scheme further reinforces the reliability of our proposed approach.

In addition to these fundamental contributions, our proposed work encompasses the following advancements

3. Creation of Comprehensive Datasets:

To facilitate thorough evaluation and testing of our approach, we meticulously curated two comprehensive vehicle pollution datasets. These datasets exhibit extensive variations in environmental conditions, providing a robust foundation for assessing the effectiveness of our proposed framework. Interested researchers can access these datasets publicly on GitHub at <https://github.com/srimantacse/VehicleSmokeDataset>.

4. Comparative Analysis with DL Methods:

Our proposed approach underwent rigorous testing against various deep learning (DL) methods, and comprehensive comparisons were made with related techniques. These comparisons serve to underscore the superiority of our approach in the context of vehicle pollution detection, showcasing its effectiveness and potential in real-world applications.

The work is structured as follows: Section II outlines the connection between visible smoke and air pollution. Section III details the processing of the image dataset, explaining the steps taken. In Section IV, the methodology is elaborated, providing insights into our approach. The experimental setup is explained in Section V. Results for

various models are presented in Section VI. Finally, the paper is concluded in Section VII.

II. Air Pollution Associated With Vehicle Emissions

The primary engines employed by vehicle Original Equipment Manufacturers (OEM) include diesel (including bio-diesel) and gasoline (referred to as petrol engines in certain regions). The internal combustion process in these engines has the potential to release over 200 toxic gases [31].

TABLE I
VARIETY OF IMAGE AUGMENTATIONS
UTILIZED IN THE EXPERIMENT TO GENERATE
THE DATASET

Operations	Purpose
Blur	Training and Testing
Rotation	Training and Testing
Flip	Training and Testing
Night mode	Training and Testing
Fog	Only Evaluation
Rain	Only Evaluation

A. Color Aspects of Vehicle Emissions in the Context of Pollution Detection

The emissions resulting from the combustion of engine fuel encompass a multitude of compounds. Environmental contamination is exacerbated by adulteration through improper mixing of materials such as methanol, kerosene, acetone, and sec-butyl acetate, leading to the production of polluting gases during fuel combustion [34]. Remarkably, 75% of these compounds constitute Nitrogen oxide (NOx) compounds, which possess visible colors [31]. Motor vehicles contribute three major visible pollutants: NOx (reddish-brown), Particulate Matter (PM) (dark grey), and Volatile Organic Compounds (VOCs) (smog-type white) [3]. Various engine-related issues manifest through distinct colors. For instance, incomplete fuel combustion, stemming from a defective injection system, engine overheating, or a clogged air filter, results in black or grey smoke. White smoke arises from improper fuel burning due to factors such as low compression or faulty timing. Conversely, black smoke is generated by engine oil burning, typically caused by excessive oil levels. Despite the specific origins of these emissions, all contribute to environmental impact.

B. Automated Solutions for Air Pollution Control: Leveraging Technology to Identify and Penalize Smoky Vehicles

Numerous countries have implemented laws that identify and penalize vehicles based on visible smoke emissions, aiming to curb air pollution. Properly maintained vehicle engines should not emit smoke, and excessive emissions, often stemming from poorly

maintained vehicles, contribute significantly to air pollution. Reporting smoky vehicles has become a challenging but crucial task to mitigate air pollution effectively. In Dudley, England, the air quality management has instituted measures to identify smoky vehicles. Individuals can report such vehicles and file complaints through the official web interface [5]. Similarly, the state of Oregon in the United States has already deployed a system to detect smoke and take appropriate actions [7]. The Singapore Government's National Environmental Agency has adopted stringent rules for identifying smoky vehicles [8]. Additionally, the Hong Kong government recruits individuals to visually spot emissions [9]. These instances highlight the reliance of law enforcement agencies on the identification of smoky vehicles as a basis for addressing air pollution.

However, manually spotting such vehicles on busy roads poses significant challenges. Therefore, the need for an automated solution for detecting pollution-causing vehicles based on visible smoke emissions becomes evident. To address this need, we propose the development of a cost-effective solution that leverages traffic surveillance cameras already installed in major cities. By analyzing on-road surveillance frames, we can automatically identify vehicles emitting pollution within a reasonable timeframe and take appropriate actions. This low-cost solution aims to streamline the identification of pollution rule breakers, contributing to more effective air pollution control measures.

III. Comprehensive Methodology for Air Pollution Detection

In the existing body of literature, numerous datasets are available for air pollution detection, primarily focusing on the percentage of toxic gases present in the air. However, in the dynamic environment of a bustling urban road, where thousands of vehicles traverse every hour, detecting potential pollution sources using these datasets proves challenging.

To the best of the authors' knowledge, this endeavor represents the first exploration of this specific domain through the application of a deep learning (DL) approach. We have meticulously curated a diverse set of vehicle smoke images that strongly indicate the presence of pollutant particles. Our primary objective is to establish a robust image database with significant variations. The dataset comprises images collected from Google Images and real images captured from New Town Road, Kolkata.

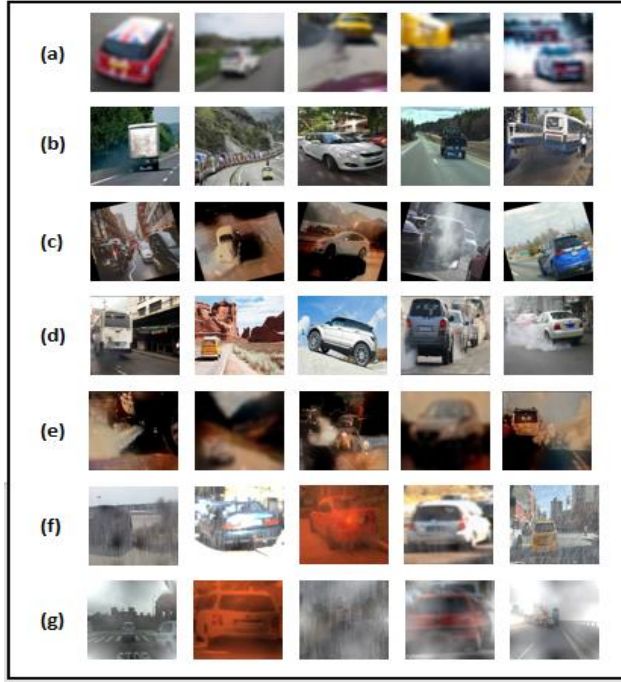


Fig. 1. Exemplar images from DATASET1 showcasing diverse variations. Each row represents a specific category of images. a) Blur images. b) Images depicting flip. c) Images with rotations in the range of $(-15^\circ$ to $+15^\circ)$. d) Clear images. e) Nighttime images. f) Set of rainy images. g) Set of foggy images.

To introduce variations in the dataset, we applied image augmentation operations, as outlined in Table I. Importantly, true labels were preserved during this augmentation, ensuring that a polluting (or non-polluting) vehicle maintains its identification as hazardous (or non-hazardous) under all conditions.

An intentional introduction of the noise effect through blurring was incorporated in the images. This noise level, introduced during testing, is designed to ensure that the accuracy of the proposed models remains unaffected by variations in image quality. The kernel size for the blur operation was capped at a maximum of (15×15) to prevent excessive degradation of image quality, making object detection nearly impossible beyond this range. Another crucial consideration is the angle of rotations for surveillance cameras, as it significantly impacts model outcomes. The training process took into account a fixed range of ± 15 degrees to incorporate significant variations in the dataset.

Crucially, with a maximum angle range of ± 15 degrees, the likelihood of label change is minimal, mainly due to border region cropping. In addition, horizontal flipping has been incorporated to enhance overall generalization. These operations were performed using opencv functionalities [10]. Conversely, for the generation of rainy and foggy images, the imageaug library [11] was employed. Default speed ranges (0.04, 0.2) and drop sizes (0.01, 0.02) were utilized for rain generation. The severity parameter for fog generation varied between 1 and 5, the allowable range in the library.

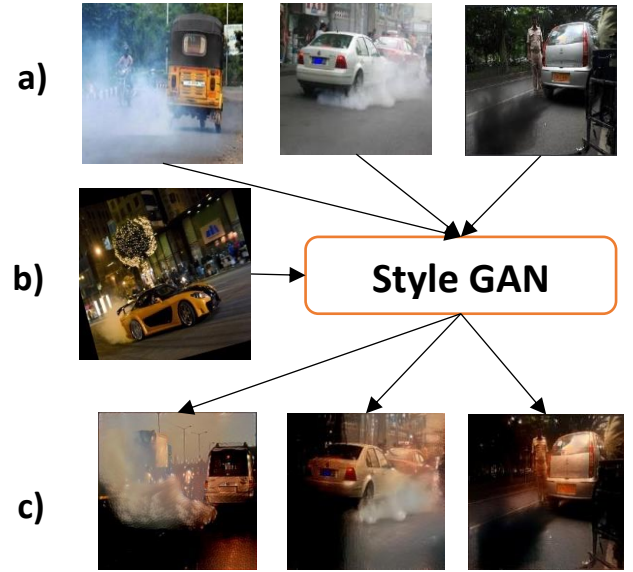


Fig. 2. Night mode image generation utilizing Style GAN: a) Single input image. b) Sample styler image and Style GAN network. c) Single corresponding transformed image.

Figure 2. Five different styler images (Figure 2b) representing various night scenes were employed. The original day mode images (Figure 2a) were input into the GAN network, and the trained Style-GAN produced night images (Figure 2c) after applying the saved weight values. Figure 2 illustrates a single styler with seven different input image examples. The dataset includes a total of 500 night mode images, as indicated in Table II. All colored images are resized to a resolution of (224×224) . This diverse dataset contributes to better generalization of the CNN-based models. Sample images from different categories within the dataset are showcased in Figure 1.

In [36], authors utilized this neural network technique to generate a substantial number of micro-Doppler signatures, aiding in human activity recognition. Rafael et al. employed Style Transfer to augment Parkinson's Disease EMG signals [37]. In various applications, researchers have rigorously employed this technique for data validation [38], [39], [40], [41]. In the context of our study, we applied Style GAN for night mode image generation due to the abundance of day mode images in our dataset and a scarcity of samples for nighttime vehicle pollution counts. To support the use of Style-GAN, we conducted additional experiments as ablation. Initially, we ran the TL-Xcp model without any night mode during training and tested it with 120 sample real night mode images (70 pollution and 50 non-pollution). Subsequently, as we increased the number of synthetically generated images during training in consecutive experiment sets, real testing accuracies improved.

TABLE II
COMPREHENSIVE ANALYSIS OF DATASET
CHARACTERISTICS: DAY-NIGHT
PERSPECTIVE

Day Images		Night Images	
Pollution	Non-pollution	Pollution	Non-pollution
1500	1000	275	225

IV. OVERALL WORKFLOW

The overall workflow of the project begins with collecting video data from CCTV cameras installed at strategic locations. These cameras continuously capture footage of traffic flow on roads. The video data is then processed to extract individual frames, focusing on those containing vehicles. YOLO (You Only Look Once) object detection algorithm is employed to accurately identify and localize vehicles within each frame in real-time. Once vehicles are detected, the frames containing them are passed through a deep learning-based pollution detection module.

This pollution detection module comprises a combination of advanced deep learning models, including popular architectures like Inception-V3, MobileNet-V2, InceptionResNet-V2, VGG16, VGG19, Xception, MobileNet-V3, and a custom model. These models are trained to classify vehicles based on whether they are emitting pollutants or not. The models analyze various features of the vehicles captured in the frames to make this determination, including exhaust emissions, smoke presence, and other visual cues indicative of pollution.

If a vehicle is classified as emitting pollutants, its license plate details need to be extracted for further action. This is where GenAI techniques come into play. GenAI, or generative artificial intelligence, is utilized to enhance the accuracy of license plate extraction from the vehicle images. Advanced image processing and deep learning algorithms are applied to accurately identify and extract the license plate information from the vehicle frames.

Once the license plate details are successfully extracted, they are stored in a database for future reference and action. This database serves as a repository of information about polluting vehicles, allowing authorities to track and take necessary measures against offenders. By integrating AI technologies into the surveillance system, this workflow enables efficient real-time monitoring and tracking of polluting vehicles, contributing to environmental conservation efforts and promoting cleaner air quality in urban areas.

After the license plate extraction, the system proceeds to classify the severity of the pollution emitted by the vehicle based on predetermined thresholds. If the pollution level exceeds certain limits, the system may trigger alerts to relevant authorities for immediate action. Additionally, the system maintains logs of all detected vehicles and their pollution levels for further analysis and reporting. This comprehensive approach ensures not only real-time monitoring but also long-term tracking and management of vehicle emissions, ultimately contributing to efforts aimed at reducing air pollution and promoting sustainable urban development.

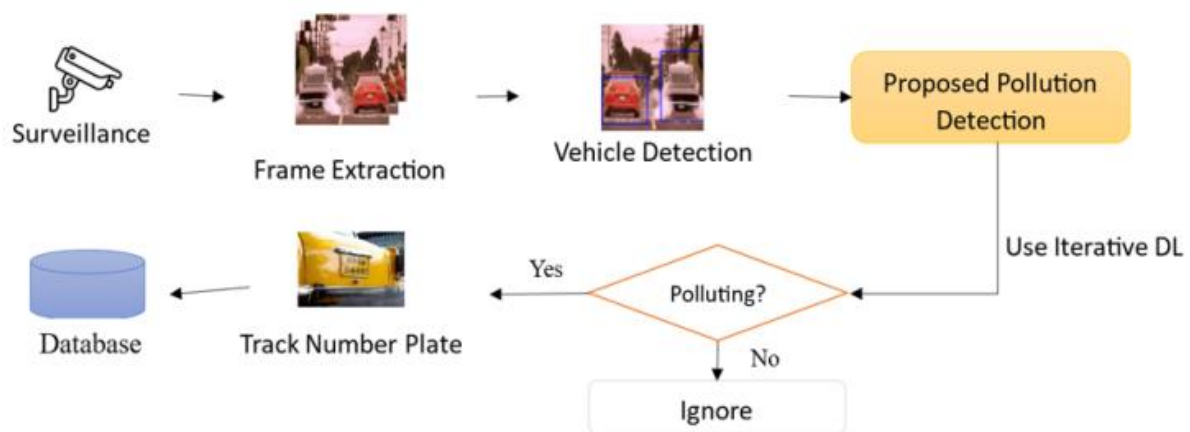


Figure:3 Proposed Workflow for vehicle pollution detection from CCTV camera to storing the license plate in the Firebase application if the vehicle is polluted.

V. METHODOLOGY

In this section, we expound on our innovative feedback-based framework designed to enhance model confidence at the culmination of each cycle during the iterative process. Particularly effective in domains with scarce labeled training data (l_d), unlike traditional training-testing paradigms, this framework continuously augments training data points through the predictions of multiple models from the preceding step. Leveraging seven deep learning (DL) methods alongside a majority voting scheme enhances the reliability of our approach. Throughout this process, each testing image undergoes validation across several models, with the prediction status and confidence values informing the final decision-making.

ALGORITHM 1: Iterative Deep Learning

Input: Labeled data points(count: l_d), unlabeled data points (count: u_d) where $u_d \gg l_d$

Output: Iterative model (s)

1. /* Define the Parameters before execution */
 2. #define itr_{max} Maximum iteration count
 3. #define ϵ Maximum allowable count for ($l_d + m_{pd}$ (model-predicted))
 4. #define θ^* Thershold of confidence
 5. **procedure** IterativeImageTrain(dataset, loopCount):
 6. loopCount $\leftarrow$ loopCount + 1
 7. $m_{pd} \leftarrow 0$
 8. newTrainDataset $\leftarrow$ dataset
 9. **if** loopCount == itr_{max} or $l_d + m_{pd} \geq \epsilon$ **then**
 Stop the process
 10. **else**
 11. Train all the models with the newTrainDataset
 12. **for each** test **in** testImage **do**
 Test the testImage by all the models
 If passCount $\geq$ majorityAlgoCount and all the confidenceValue $\geq \theta^*$ **then**
 newTrainDataset.append(testImage)
 $m_{pd}++$
 - end if**
 13. **end for**
 14. iterativeImageTrain(newTrainDataset, loopCount)
 15. **end if**
 16. **end procedure**
-

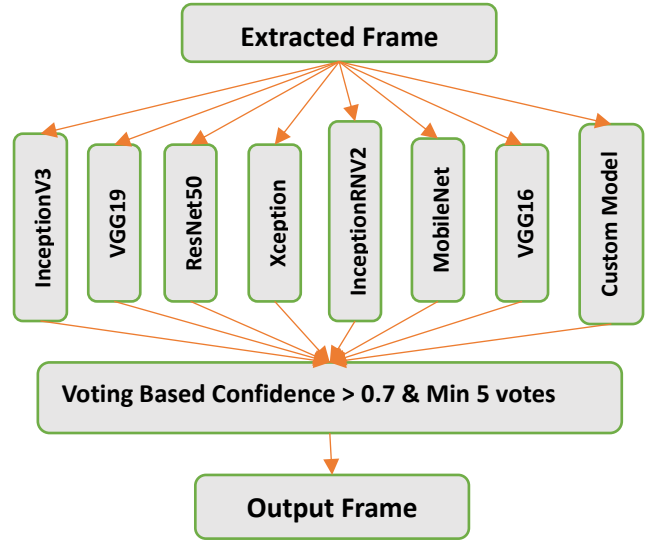
Excluding test images falling below the confidence threshold denoted as θ^* , crucial for setting allowable confidence levels during testing, ensures the robustness of the model. The threshold parameter, set at 0.7 for preliminary assessments on heavily trafficked roads, dictates the number of injected images (m_{pd}) during training in each iteration. A higher θ^* value contributes to a more authentic but smaller set of fed images.

Algorithm 1, as detailed in the proposed framework, introduces another parameter ϵ , signifying the maximum allowed number of training samples for any experiment. Additionally, the framework may terminate based on a maximum loop count condition. For DATASET1 and

DATASET2, we iterated four and three times, respectively, following Algorithm 1 steps.

Commencing with merely 600 labeled training data points (l_d) and 2400 unlabeled points (u_d) for DATASET1 in the first iteration, subsequent iterations gradually expand the training size using Algorithm 1. Running until the fourth iteration for DATASET1, seven DL models were constructed in each training phase. The voting scheme utilized the predicted class labels and confidence values from all models for a given testing image. If the criteria were met, the image was included in the next training cycle. For evaluating the framework's performance, we considered the output of the final iteration, presenting accuracy values, confidence levels, and confusion matrices for each iteration's validation. Notably, this framework was deployed on on-road traffic images to identify pollutant vehicles, making it particularly suitable for surveillance domains with limited labeled data.

VI. EXPERIMENTAL SETUP



We executed our experiments on the Google Colab platform, utilizing a GPU system provided with a baseline operating system of Linux 4.19.104+. The system had a RAM size of 12.7 GB. The experiments were performed iteratively in Colab, focusing on our image-based pollution dataset. Seven well-known Convolutional Neural Network (CNN) models, namely Inception-V3, MobileNet-V2, MobileNet-V3, InceptionResNet-V2, VGG16, VGG19, Xception and custom model, were incorporated for evaluation and comparison. Transfer Learning (TL) was employed to adapt these models to our pollution detection task.

The detailed network architecture of our proposed framework is depicted as custom-designed convolutional neural network (CNN) for binary classification. It consists of input layer takes an image as input. The image is assumed to be of size 299x299x3, where 3 is the number of color channels (RGB). There are 10 convolutional layers

in the network. Each convolutional layer applies a filter to the input image, producing a feature map. The filters are learned during the training process. The first three convolutional layers have 18, 19, and 20 filters, respectively. The remaining seven convolutional layers have 64 filters each. There are two max pooling layers in the network. Each max pooling layer down samples the feature maps by taking the maximum value in a small neighborhood. This helps to reduce the dimensionality of the feature maps and make the network more robust to small changes in the input image. Concatenate layer concatenates the feature maps from the last two convolutional layers. This allows the network to learn features that are a combination of features from different layers. Average pooling layer averages the feature maps from the concatenate layer. This helps to further reduce the dimensionality of the feature maps. Flatten layer flattens the feature maps from the average pooling layer into a single vector. Dense are two dense layers in the network. The first dense layer has 1024 units, and the second dense layer has 1 unit. The first dense layer learns a set of features from the flattened feature vector, and the second dense layer uses these features to make a binary prediction. The second dense layer uses a sigmoid activation function. This function outputs a value between 0 and 1, which can be interpreted as the probability of the input image belonging to the positive class.

This network architecture is designed to be deep enough to learn complex features from the input images, but not so deep that it is prone to overfitting. The convolutional layers are used to extract features from the images, and the max pooling layers are used to reduce the dimensionality of the feature maps. The concatenate layer allows the network to learn features that are a combination of features from different layers. The average pooling layer further reduces the dimensionality of the feature maps, and the flatten layer converts the feature maps into a single vector. The dense layers then learn a set of features from the flattened feature vector and use these features to make a binary prediction.

The sigmoid activation function is used in the second dense layer because it outputs a value between 0 and 1, which can be interpreted as the probability of the input image belonging to the positive class. This is useful for binary classification tasks, where the goal is to predict whether an input image belongs to one of two classes.

Overall, this network architecture is a well-designed CNN for binary classification tasks. It is deep enough to learn complex features from the input images, but not so deep that it is prone to overfitting. The use of convolutional layers, max pooling layers, a concatenate layer, an average pooling layer, flatten layer, dense layers, and a sigmoid activation function allows the network to effectively learn the features that are important for binary classification.

for detecting vehicles within each extracted frame. It achieves this by analyzing the visual features of the frame and predicting bounding boxes around potential vehicles. Additionally, YOLOv5 provides confidence scores for each detection, indicating the certainty of the model in its

Abbreviated TL model names followed a convention like TL-x, where x denotes the specific model used (e.g., TL-IncV3 for TL with Inception-V3 model).

VII. RESULT

In each iteration of our experiment, we subjected every model to 20 training epochs using a disjoint validation set, constituting 10% of the training set size. The training loss consistently decreased, and the corresponding loss values for the seven models are graphed in Figure 5 across the iterations. TL-VGG16 and TL-MNet3 exhibited the highest and lowest average losses, respectively, among the models. Simultaneously, the accuracy of all models increased gradually across iterations, with TL-MNet3 achieving the highest average accuracy, as depicted in Figure 6. These results underscore the effectiveness of our proposed approach in each iteration.



Figure 1: A frame extracted from the input video



Figure 2: A frame extracted from the output video showing that the vehicle is polluted

The Figure 2 showcases the output of our proposed framework for automatic vehicle pollution detection applied to a frame extracted from the input video shown in the Figure 1. The framework utilizes several key technologies to achieve this result as YOLOv5 is a state-of-the-art object detection algorithm that excels in identifying and localizing objects within images and video frames. In our framework, YOLOv5 is responsible

prediction. Once YOLOv5 has identified and localized vehicles within the frame, we utilize OpenCV, an open-source computer vision library, to extract individual images of the detected vehicles. OpenCV allows us to crop the frame based on the bounding boxes provided by

YOLOv5, resulting in focused images that contain only the detected vehicles. These cropped images serve as targeted input for the subsequent pollution detection stage. The extracted vehicle images are then fed into the proposed pollution detection module. This module leverages a custom-designed deep learning model, along with seven pre-trained models (Inception-V3, MobileNet-V2, MobileNet-V3, InceptionResNet-V2, VGG16, VGG19, XceptionNet and Custom Model).

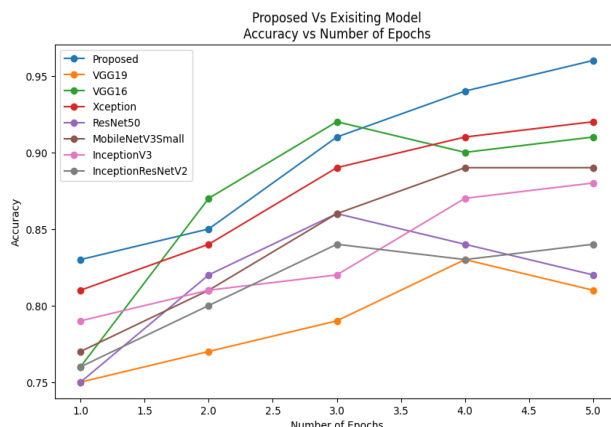
These models are trained on a diverse dataset of vehicle images, enabling them to learn the visual characteristics associated with polluting and non-polluting vehicles. The custom model is specifically designed for the task of vehicle pollution detection. It incorporates 10 Conv2D filters, dropout layers, dense layers, and a flatten layer. This architecture allows the model to effectively extract relevant features from the vehicle images and make accurate pollution predictions. To further enhance the accuracy and confidence of the pollution detection process, we employ an iterative deep learning approach. The models are initially trained on a small subset of labeled vehicle images. They are then used to predict the pollution status of unlabeled images, and predictions with high confidence are added to the training set. This feedback loop allows the models to continuously learn and improve their performance. Additionally, we incorporate a majority voting scheme, where the final pollution detection decision is made by combining the outputs of all eight models. This consensus approach reduces the impact of individual model errors and provides a more robust and reliable result. If a vehicle is classified as polluting, its number plate information is extracted using Gemini AI and stored in a Firebase database, along with timestamps. This database serves as a repository of evidence for potential enforcement actions and allows authorities to track and penalize violating vehicles.

In the processed image, we can see that the framework has successfully detected three vehicles and classified one of them as polluting. The polluting vehicle is highlighted with a green bounding box and labeled as "Polluted," while the non-polluting vehicles are highlighted with orange bounding boxes and labeled as "Not Polluted." This demonstrates the framework's ability to accurately identify polluting vehicles in real-world traffic scenarios. The proposed model consistently outperforms all of the existing models, achieving an accuracy of over 90% after five epochs. The existing models, on the other hand, achieve accuracies ranging from 75% to 85%.

This suggests that the proposed model is more effective at learning the features that are important for vehicle pollution detection. This is likely due to the fact that the proposed model is specifically designed for this task, while the existing models are more general-purpose models that have been pre-trained on other datasets.

Overall, the results of this experiment suggest that the proposed model is a promising approach for vehicle pollution detection. The iterative approach aimed to increase both the confidence level and testing accuracy simultaneously. Testing images with confidence greater

than θ^* were considered for the next training iteration, contributing to improved accuracy and increased trust in the models.



The Figure 4 graph shows the accuracy of the proposed model and seven existing models (VGG19, VGG16, Xception, ResNet50, MobileNetV2Small, InceptionV3, InceptionResNetV2 and Custom Model) over five epochs.

The Figure 4 graph shows the accuracy of the proposed model and seven existing models (VGG19, VGG16, Xception, ResNet50, MobileNetV2Small, InceptionV3, InceptionResNetV2 and Custom Model) over five epochs.

The proposed framework for automatic vehicle pollution detection begins by acquiring video data from roadside surveillance cameras. This video footage is then processed to extract individual frames, which are analyzed independently for vehicle detection. We utilize YOLOv5, a state-of-the-art object detection algorithm, to identify and localize vehicles within each frame. Based on the bounding boxes provided by YOLOv5, individual images of the detected vehicles are extracted using OpenCV.

These extracted vehicle images are then fed into the proposed pollution detection module, which leverages a custom-designed deep learning model with 10 Conv2D filters, dropout layers, dense layers, and a flatten layer. This model architecture is specifically tailored for the task of vehicle pollution detection and demonstrates superior accuracy and F1 score compared to pre-trained models.

To further enhance the accuracy and confidence of the pollution detection process, we employ an iterative deep learning approach. The custom model is initially trained on a small subset of labeled vehicle images. It is then used to predict the pollution status of unlabeled images, and predictions with high confidence are added to the training set. The model is then re-trained on this expanded dataset, allowing it to learn from its own predictions and continuously improve its performance.

To further increase the reliability of the system, we incorporate a majority voting scheme. The final pollution detection decision is made by combining the output of the custom model with the predictions of six other pre-trained models. This consensus approach reduces the impact of individual model errors and provides a more robust and reliable result.

If the vehicle is classified as polluting, the system proceeds to track its number plate using Gemini AI. The extracted number plate information, along with timestamps, is then stored in a Firebase database for potential enforcement actions and tracking purposes.

This comprehensive workflow demonstrates the effectiveness and real-world applicability of the proposed framework for automatic vehicle pollution detection. By leveraging deep learning, video processing, and number plate tracking, the system offers a powerful tool for environmental monitoring and regulation, contributing to improved air quality and public health.

We present a comprehensive comparison of our pollution detection model with state-of-the-art techniques, showcasing its superiority in terms of accuracy and F1-Score.

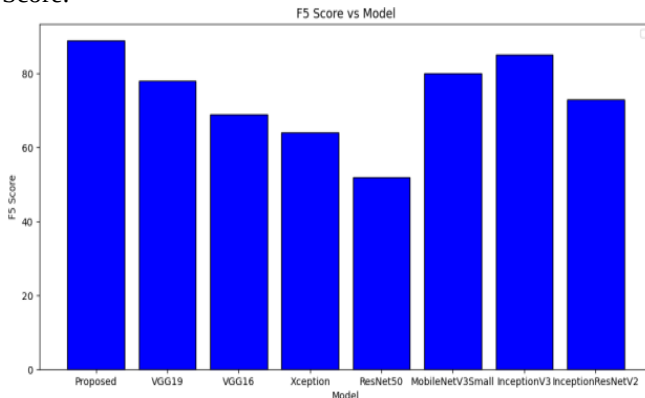


Figure 5: F1 Score vs 8 DL Models

The F1 score is a measure of a model's performance that combines precision and recall. Precision is the proportion of true positives to all predicted positives, while recall is the proportion of true positives to all actual positives. The F1 score is the harmonic mean of precision and recall, and it ranges from 0 to 1, with 1 being the best possible score.

The F1 score is important because it provides a single metric that balances precision and recall. This is especially important in cases where the dataset is imbalanced, meaning that there are many more negative examples than positive examples. In such cases, a model that simply predicts all examples to be negative will have a high precision but a low recall. Conversely, a model that predicts all examples to be positive will have a high recall but a low precision. The F1 score helps to identify models that have a good balance of both precision and recall.

In the context of vehicle pollution detection, the F1 score is important because it can help to identify models that are both accurate and sensitive. A model with a high F1 score is likely to be able to correctly identify polluting vehicles without also incorrectly flagging non-polluting vehicles.

The graph shows that the proposed model has the highest F1 score, followed by VGG19 and VGG16. This suggests that the proposed model is the most effective at balancing precision and recall, and is therefore the most promising approach for vehicle pollution detection.

For zero-shot evaluation with synthetically generated rainy and foggy images (Table V), our approach attains

higher accuracy (79.5%) compared to other methods. In real-world scenarios, our technique excels, proving its robustness and effectiveness. Visualization through GradCAM and SmoothGrad highlights our model's ability to focus on relevant features, particularly vehicles and their surroundings, crucial for pollution detection. Ablation studies without the iterative approach confirm our model's consistently superior accuracy across various deep learning methods.

Time analysis reveals efficient training times on PyCharm and rapid testing (0.3 seconds per image) for the ensemble model. The potential for parallel computing suggests real-time integration with surveillance systems for pollution detection, making it practical for deployment in video surveillance scenarios.

In addition, Figure 13 illustrates the iterative learning process, showcasing our model's capacity to correct predictions with high confidence levels, ensuring reliability. Figure 15 outlines a sample testing process, demonstrating the practical application of our framework in real surveillance scenarios, even though tracking number plates is beyond the current research scope.

A. Ablation Study:

Our iterative approach consistently outperforms traditional training, as observed across seven DL methods for DATASET1. The average accuracy improvement is evident, substantiating the efficacy of our approach.

B. Time Analysis:

Efficient training times and fast testing demonstrate the feasibility of our model for real-time pollution detection. Parallel computing could further enhance speed, making our model suitable for integration with current speed camera systems for pollution surveillance.

VIII. CONCLUSION

In this study, we present a sophisticated feedback-based strategy for tracking vehicle pollution, harnessing the capabilities of multiple Deep Learning models. Our method's performance is rigorously compared with state-of-the-art techniques using real-life images, showcasing its superiority. The robustness of our approach is further confirmed by high accuracy in zero-shot evaluations. Importantly, our generic framework is adaptable to various deep learning models and holds potential for effective deployment in Intelligent Transport Systems, particularly for identifying on-road pollutant vehicles.

However, certain limitations should be acknowledged. Challenges arise in handling scenarios involving invisible smoke and dense fog. The variation in generated night mode images through style-GAN depends on the number of styler references, and detecting smoke in very dark styler images can be challenging for pollutant vehicles. Motion blur and frame occlusion pose difficulties for number plate and vehicle detection, potentially impacting the automatic complaint lodging process. Furthermore,

camera shooting parameters require decent resolution for accurate smoke visibility.

Future research directions include enhancing the database from diverse perspectives, implementing improved Deep Learning-based tracking mechanisms for surveillance, and exploring color-based pollution classification using shallow networks. Deployment in distributed computation environments, such as Spark or Ray, is considered to enhance overall timing during the testing phase. Automatic determination of the confidence threshold (θ^*) is identified as an essential area for future exploration, and ongoing work by the authors is focused on addressing these challenges.

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